Name: _____

TOPIC TEST 2

Chapter 12: Applications of derivatives

- Time allowed: 45 minutes
- CAS calculators may be used.
- Part A: 10 multiple-choice questions (10 marks)
- Part B: 8 short answer questions (40 marks)
- Part C: 5 analysis questions (30 marks)

Part A

10 multiple-choice questions
1 mark each: 10 marks
Circle the correct answer.

1 $f(x) = 3x^2 - 5x + 13$. The instantaneous rate of change when $x = 2$ is:

A -17

B 7

C 17

D -5

E 15

[1 mark]

2 The values of a function and its derivative for selected points are given in the table. All values for which the derivative is zero are shown. The function has no asymptotes.

x	-4	-2	0	1	3
$f(x)$		-6		8	
$f'(x)$	-5	0	6	0	-4

The function has:

A a minimum at (1, 8)

B a maximum at $(-2, 0)$

C a point of inflection at $(-2, 0)$

D a maximum at (1, 8)

E a point of inflection at (0, 6).

[1 mark]

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3 The function $f(x)$ has a derivative that can be factorised to $f'(x) = (x - 3)(5 + x)$. $f(x)$ has:

A a minimum at $x = 3$ and a maximum at $x = -5$

B a maximum at $x = 3$ and a minimum at $x = -5$

C a minimum at $x = -3$ and a maximum at $x = 5$

D a maximum at $x = -3$ and a minimum at $x = 5$

E points of inflection at $x = 3$ and $x = -5$.

[1 mark]

4 The function $f(x) = x^3 - 5x^2 + 3x + 12$ has a:

A minimum at $\left(-\frac{1}{3}, 12\frac{1}{2}\right)$

B maximum at $\left(\frac{1}{3}, -12\frac{1}{2}\right)$

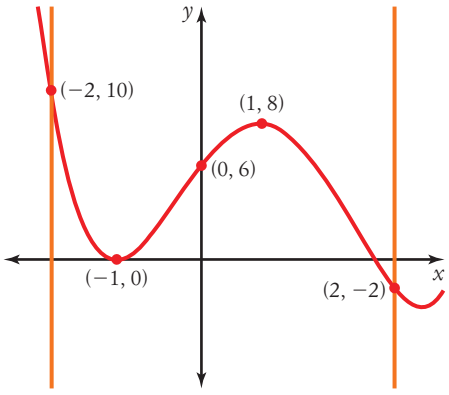
C minimum at $(3, -3)$

D maximum at $(3, 3)$

E maximum at $\left(\frac{1}{3}, 12\frac{1}{2}\right)$.

[1 mark]

The diagram below shows the graph of a function, $f(x)$, over the domain $-2 \leq x \leq 2$. Use the graph to answer questions **5** to **10**.



5 A zero for this function can be found at:

A $(1, 8)$

B $(-1, 0)$

C $(2, -2)$

D $(0, 6)$

E $(-2, 10)$

[1 mark]

6 A local minimum for this function can be found at:

A $(1, 8)$

B $(-1, 0)$

C $(2, -2)$

D $(0, 6)$

E $(-2, 10)$

[1 mark]

7 The greatest value on the specified domain can be found at:

A $(1, 8)$

B $(-1, 0)$

C $(2, -2)$

D $(0, 6)$

E $(-2, 10)$

[1 mark]

8 The least value on the specified domain can be found at:

A $(1, 8)$

B $(-1, 0)$

C $(2, -2)$

D $(0, 6)$

E $(-2, 10)$

[1 mark]

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9 A local maximum for this function can be found at:

A $(1, 8)$

B $(-1, 0)$

C $(2, -2)$

D $(0, 6)$

E $(-2, 10)$

[1 mark]

10 The average rate of change between $(1, 8)$ and $(2, -2)$ is:

A -3

B -10

C $\frac{1}{10}$

D $-\frac{3}{2}$

E 6

[1 mark]

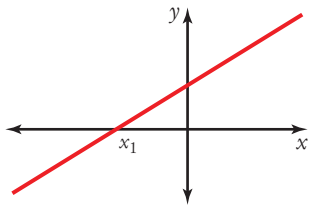
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Part B

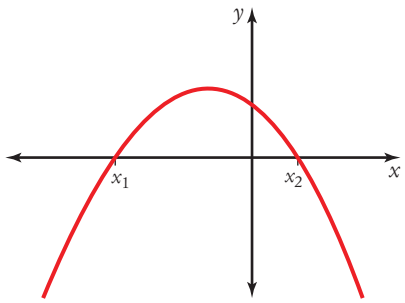
8 short answer questions
40 marks
Show your working where appropriate.

1 Sketch a primitive function for each graph.

a



b



[9 marks]

2 The velocity of a particle is given by $V = 6t - 12t^2$ ms^{-1}.
Find its displacement and acceleration after
3 seconds if it is initially 5 m to the right of the origin.

[5 marks]

3 A particle's displacement, at time t seconds, is modelled by the equation $x = t^3 - 12t^2 + 36t - 9$, where x is in cm.

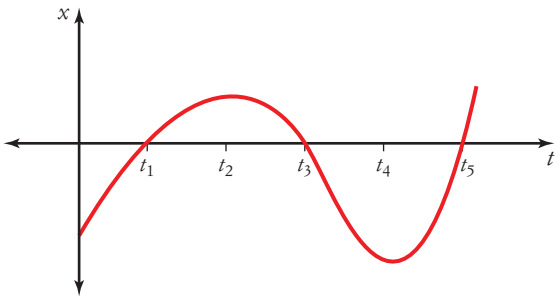
a When is the particle at rest?

b What is the displacement, velocity and acceleration after 1 s?

c Describe the motion of the particle after 1 s.

[7 marks]

4 The graph below shows the displacement of a particle over time t.



a When is the particle at the origin?

b When is the particle at rest?

[2 marks]

5 Find the stationary point(s) on the curve $y = 2x^2 + 7$ and determine its nature.

[3 marks]

6 Show that $y = -x^3$ is always decreasing for all $x \neq 0$.

[3 marks]

7 Find all values of x for which the curve $y = 4x^3 - 21x^2 - 24x + 5$ is increasing.

[5 marks]

8 Find the greatest and least values of the function $f(x) = 4x^3 - 3x^2 - 18x$ in the domain $-2 \leq x \leq 3$.

[6 marks]

Part C

5 analysis questions
30 marks
Show your working where appropriate.

1 Water is flowing out of a pool at the rate given by $R = 20t$ litres per minute. If the volume of water in the pool is initially 8000 L, find:

a the volume after 5 minutes

b how long it will take to empty the pool.

[8 marks]

2 The acceleration of a particle is given by $a = 15t + 12$ m/s^2. When $t = 2$, the velocity $= 5$ m/s and displacement $= 5$ m. Find an equation for the displacement of the particle.

[4 marks]

3 Identify the stationary points of the function $f(x) = x^4 - 12x^3 - 4$.

[6 marks]

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4 Sketch the curve of the function $y = 3x^4 + 8x^3 - 48x^2 + 1$.

[6 marks]

5 A gardener wishes to make a rectangular garden bed using a wall as one of the sides. She has 24 m of edging strip to place around the garden. What dimensions will give the maximum garden area?

[6 marks]
Total marks: 80

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Answers

Part A [1 mark each]

1 $f'(x) = 6x - 5$

$f'(2) = 6(2) - 5 = 7$

$\therefore$ **B** **[1 mark]**

2 When $x < -2, f'(x) < 0$, when $x = -2, f'(x) = 0$ and when $x > -4, f'(x) > 0$. Minimum at -2.

When $x < 1, f'(x) > 0$, when $x = 1, f'(x) = 0$ and when $x > 1, f'(x) < 0$. Maximum at 1.

$\therefore$ **D** **[1 mark]**

3 Since its derivative function is a quadratic, $f(x)$ must be a cubic.

Let $f'(x) = 0$.

$(x - 3)(5 + x) = 0$

$\therefore x = 3, -5$

To determine the nature of these zeros, find the value of $f'(x)$ on either side of $x = 3$ and $x = -5$.

When $x < 3, f'(x) < 0$, when $x = 3, f'(x) = 0$ and when $x > 3, f'(x) > 0$. Minimum at 3.

When $x < -5, f'(x) > 0$, when $x = -5, f'(x) = 0$ and when $x > -5, f'(x) < 0$. Maximum at -5.

$\therefore$ **A** **[1 mark]**

4 $f'(x) = 3x^2 - 10x + 3 = (3x - 1)(x - 3)$

Let $f'(x) = 0$

$(3x - 1)(x - 3) = 0$

$\therefore x = \frac{1}{3}, 3$

To determine the nature of these zeros, find the value of $f'(x)$ on either side of $x = 3$ and $x = \frac{1}{3}$.

When $x < 3, f'(x) < 0$, when $x = 3, f'(x) = 0$ and when $x > 3, f'(x) > 0$. Minimum at 3.

$f(3) = (3)^3 - 5(3)^2 + 3(3) + 12 = 27 - 45 + 9 + 12 = 3$

When $x < \frac{1}{3}, f'(x) > 0$, when $x = \frac{1}{3}, f'(x) = 0$ and when $x > \frac{1}{3}, f'(x) < 0$. Maximum at $\frac{1}{3}$.

$f\left(\frac{1}{3}\right) = \left(\frac{1}{3}\right)^3 - 5\left(\frac{1}{3}\right)^2 + 3\left(\frac{1}{3}\right) + 12 = \frac{1}{27} - \frac{5}{9} + 1 + 12 = 12\frac{13}{27}$

$\therefore$ **E** **[1 mark]**

5 Of the options given, only **B** has a zero y-value.

$\therefore$ **B** **[1 mark]**

6 Over the given domain, the local minimum occurs at $(-1, 0)$.

$\therefore$ **B** **[1 mark]**

7 Over the given domain, the maximum value of the function occurs at $(-2, 10)$.

$\therefore$ **E** **[1 mark]**

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8 Over the given domain, the minimum value of the function occurs at $(2, -2)$.

$\therefore$ **C** **[1 mark]**

9 Over the given domain, the local maximum occurs at $(1, 8)$.

$\therefore$ **A** **[1 mark]**

10 Average rate of change $= \dfrac{\text{Rise}}{\text{Run}}$

$$= \frac{-2 - 8}{2 - 1}$$
$$= -10$$

$\therefore$ **B** **[1 mark]**

Part B

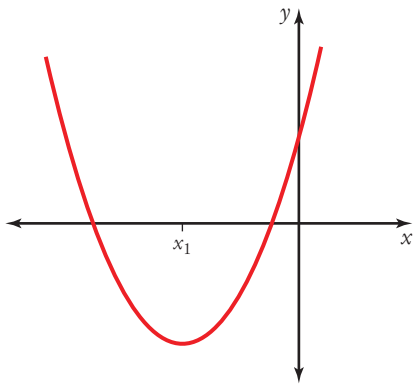
1 a Draw a set of axes and mark in point x_1 in the same position on the x-axis.

The primitive function will have a stationary point at x_1. **[1 mark]**

The primitive function will have:

- a negative slope to the left of x_1 **[1 mark]**
- a positive slope to the right of x_1 **[1 mark]**

Draw the graph. **[1 mark]**



b Draw a set of axes and mark in points x_1 and x_2 in the same positions on the x-axis.

The primitive function will have stationary points at x_1 and x_2. **[1 mark]**

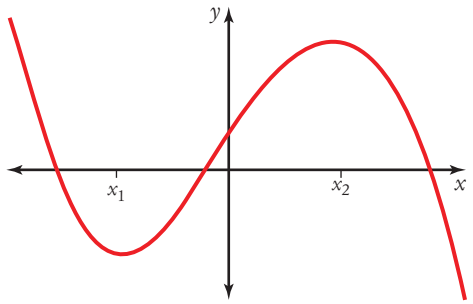
The primitive function will have:

- a negative slope from the left up to x_1 **[1 mark]**
- a positive slope between x_1 and x_2 **[1 mark]**
- a negative slope from x_2 and beyond. **[1 mark]**

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Draw the graph. **[1 mark]**

[9 marks]



2 Displacement $= s = \frac{6t^2}{2} - \frac{12t^3}{3} + c = 3t^2 - 4t^3 + c$ **[1 mark]**

When $t = 0, s = 5$

$5 = 3 \times 0^2 - 4 \times 0^3 + c$

$\therefore c = 5$

$\therefore s = 3t^2 - 4t^3 + 5$ **[1 mark]**

Acceleration $= a = \frac{dV}{dt} = 6 - 24t$ **[1 mark]**

Let $t = 3$.

$s = 3 \times 3^2 - 4 \times 3^3 + 5 = -76$ m **[1 mark]**

$a = 6 - 24 \times 3 = -66\ \text{ms}^{-2}$ **[1 mark]**

[5 marks]

3 a The particle is at rest when the velocity, $v = 0$

$v = \frac{dx}{dt} = 3t^2 - 24t + 36$

$= 3(t - 6)(t - 2)$ **[1 mark]**

Let $v = 0$

$3(t - 6)(t - 2) = 0$

$\therefore t = 2$ or 6

The particle is at rest at $t = 2$ s or 6 s. **[1 mark]**

b Acceleration $= a = \frac{dv}{dt} = 6t - 24$ **[1 mark]**

When $t = 1$:

$x = (1)^3 - 12 \times (1)^2 + 36 \times 1 - 9 = 16$ cm **[1 mark]**

$v = 3 \times (1)^2 - 24 \times 1 + 36 = 15\ \text{cms}^{-1}$ **[1 mark]**

$a = \frac{dv}{dt} = 6 \times 1 - 24 = -18\ \text{cms}^{-2}$ **[1 mark]**

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c The particle is 16 cm to the right of the origin, travelling at 15 cms^{-1} to the right. Acceleration is -18 cms^{-2} (to the left), so the particle is slowing down. **[1 mark]**

[7 marks]

4 a The particle is at the origin when the graph crosses the t-axis. This occurs at t_1, t_3 and t_5. **[1 mark]**

b The particle is at rest when its velocity is 0. This occurs at the stationary points at t_2 and t_4. **[1 mark]**

[2 marks]

5 $\frac{dy}{dx} = 4x$

Let $\frac{dy}{dx} = 0$

$4x = 0$

$\therefore x = 0$ **[1 mark]**

When $x = 0$, $y = 7$ so there is a stationary point at $(0, 7)$. **[1 mark]**

When $x < 0$, $y' < 0$; $x = 0$, $y' = 0$; $x > 0$, $y' > 0$, i.e. a local minimum at $(0, 7)$. **[1 mark]**

[3 marks]

6 For a decreasing curve, $\frac{dy}{dx} < 0$ **[1 mark]**

$\frac{dy}{dx} = -3x^2 < 0$ (since $x^2 > 0$ for all $x \neq 0$) **[1 mark]**

so the function is decreasing for all $x \neq 0$. **[1 mark]**

[3 marks]

7 For an increasing curve, $\frac{dy}{dx} > 0$ **[1 mark]**

$\frac{dy}{dx} = 12x^2 - 42x - 24 = 6(x - 4)(2x + 1)$

Let $\frac{dy}{dx} = 0$

$6(x - 4)(2x + 1) = 0$

$\therefore x = 4$ or $x = -\frac{1}{2}$ **[1 mark]**

When $x < 4$, $\frac{dy}{dx} < 0$ and when $x > 4$, $\frac{dy}{dx} > 0$ **[1 mark]**

When $x < -\frac{1}{2}$, $\frac{dy}{dx} > 0$ and when $x > -\frac{1}{2}$, $\frac{dy}{dx} < 0$ **[1 mark]**

The curve is increasing when $x < -\frac{1}{2}$ and $x > 4$. **[1 mark]**

[5 marks]

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8 $f'(x) = 12x^2 - 6x - 18 = 6(2x - 3)(x + 1)$ **[1 mark]**

Let $f'(x) = 0$

$6(2x - 3)(x + 1) = 0$

$\therefore x = -1 \text{ or } 1\frac{1}{2}$ **[1 mark]**

When $x < -1, f'(x) > 0; x = -1, f'(x) = 0; x > -1 \left(\text{but} < \frac{1}{2}\right), f'(x) < 0$, i.e. a local maximum.

[1 mark]

When $x = -1, f(x) = 11$, so there is a local maximum at $(-1, 11)$.

When $x < 1\frac{1}{2}$ (but > -1), $f'(x) < 0; x = 1\frac{1}{2}, f'(x) = 0; x > 1\frac{1}{2}, f'(x) > 0$, i.e. a local minimum.

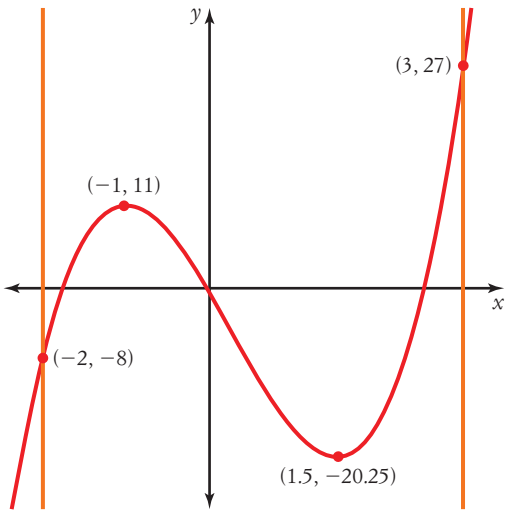
[1 mark]

When $x = 1\frac{1}{2}, f(x) = -20.25$, so there is a local minimum at $\left(1\frac{1}{2}, -20.25\right)$.

When $x = -2, f(x) = 4 \times (-2)^3 - 3 \times (-2)^2 - 18 \times (-2) = 0$

When $x = 3, f(x) = 4 \times (3)^3 - 3 \times (3)^2 - 18 \times (3) = 27$ **[1 mark (for both boundary values)]**

Draw a diagram.



Greatest value = 27 and least value = −20.25 **[1 mark]**

[6 marks]

Part C

1 a Water is flowing out, so the rate is negative.

$R = \frac{dV}{dt} = -20t$ **[1 mark]**

$V = -\frac{20t^2}{2} + c = -10t^2 + c$ **[1 mark]**

When $t = 0$, $V = 8000$

$\therefore -10t^2 + c = 8000$ **[1 mark]**

$c = 8000$

$\therefore V = -10t^2 + 8000$ **[1 mark]**

Let $t = 5$

$V = -10 \times 5^2 + 8000 = 7750$ L **[1 mark]**

b Let $V = 0$.

$-10t^2 + 8000 = 0$ **[1 mark]**

$t^2 = \frac{8000}{10} = 800$ **[1 mark]**

$t \approx 28.28$ **[1 mark]**

[8 marks]

2 Velocity is the antiderivative of acceleration so $v = \frac{15t^2}{2} + 12t + c$ **[1 mark]**

When $t = 2$, $v = 5$

$\Rightarrow \frac{15 \times 2^2}{2} + 12 \times 2 + c = 5$

$\therefore c = -49$

$\therefore v = \frac{15t^2}{2} + 12t - 49$ **[1 mark]**

Displacement is the antiderivative of velocity so

$s = \frac{15t^3}{6} + \frac{12t^2}{2} - 49t + d$

$= \frac{5t^3}{2} + 6t^2 - 49t + d$ **[1 mark]**

When $t = 2$, $s = 5$

$\frac{5 \times 2^3}{2} + 6 \times 2^2 - 49 \times 2 + d = 5$

$\therefore d = 59$

Displacement $= s = \frac{5t^3}{2} + 6t^2 - 49t + 59$ **[1 mark]**

[4 marks]

3 $f'(x) = 4x^3 - 36x^2 = 4x^2(x - 9)$ **[1 mark]**

Let $f'(x) = 0$

$4x^2(x - 9) = 0$

$\therefore x = 0$ or 9 **[1 mark]**

When $x < 0, f'(x) < 0; x = 0, f'(x) = 0; x > 0$ (but < 9), $f'(x) < 0$, i.e. a point of inflection on a falling curve. **[1 mark]**

When $x = 0, f(x) = -4$, so there is a point of inflection at, $(0, -4)$. **[1 mark]**

When $x < 9$ (but > 0), $f'(x) < 0; x = 9, f'(x) = 0; x > 9, f'(x) > 0$, i.e. a local minimum **[1 mark]**

When $x = 9, f(x) = 9^4 - 12 \times 9^3 - 4 = -2191$, so there is a local minimum at $(9, -2191)$.

Point of inflection on the falling curve is $(0, -4)$; minimum is $(9, -2191)$. **[1 mark]**

[6 marks]

4 $\frac{dy}{dx} = 12x^3 + 24x^2 - 96x$

Let $\frac{dy}{dx} = 0$

$12x^3 + 24x^2 - 96x = 12x(x - 2)(x + 4) = 0$ **[1 mark]**

$\therefore x = -4, 0$ or 2 **[1 mark]**

Let $x = -4$

$y = 3 \times (-4)^4 + 8 \times (-4)^3 - 48 \times (-4)^2 = -511$

So there is a stationary point at $(-4, -511)$.

Let $x = 0$

$y = 1$, so there is a stationary point at $(0, 1)$.

Let $x = 2$

$y = 3 \times (2)^4 + 8 \times (2)^3 - 48 \times (2)^2 = -79$

So there is a stationary point at $(2, -79)$. **[1 mark (for all three stationary points)]**

When $x < -4, y' < 0; x = -4, y' = 0; x > -4$ (but < 0), $y' > 0$, i.e. a local minimum at $(-4, -511)$.

When $x < 0$ (but -4), $y' > 0; x = 0, y' = 0; x > 0$ (but < 2), $y' < 0$, i.e. a local maximum at $(0, 1)$.

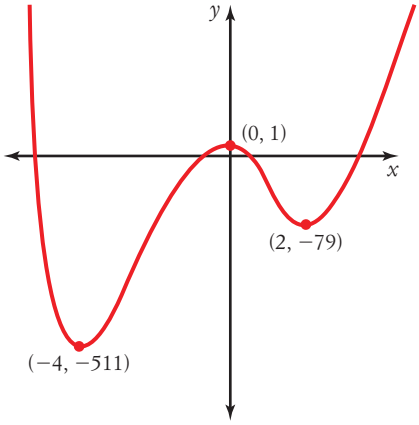
When $x < 2$ (but > 0), $y' < 0; x = 2, y' = 0; x > 2, y' > 0$, i.e. a local minimum at $(2, -79)$.

[1 mark (for identifying all three stationary points)]

$y = 3x^4 + 8x^3 - 48x^2 + 1$, so $3x^4$ is the dominant term.

As $x \to \infty, y \to \infty$ and as $x \to -\infty, y \to \infty$ **[1 mark]**

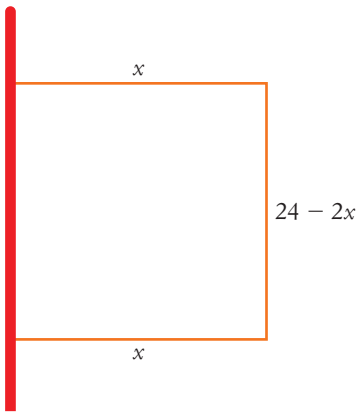
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[1 mark (for the sketch – omit axis ticks and labels)]

[6 marks]

5 Draw a diagram. Let the sides of the garden bed that are perpendicular to the wall be x.
The third side must be $(24 - 2x)$. **[1 mark]**



$A = x(24 - 2x) = 24x - 2x^2$ **[1 mark]**

$A' = 24 - 4x$ **[1 mark]**

Let $A' = 0$

$24 - 4x = 0$

$\therefore x = 6$ **[1 mark]**

When $x < 6$, $A' > 0$; $x = 6$, $A' = 0$; $x > 6$, $A' < 0$, i.e. a local maximum when $x = 6$. **[1 mark]**

When $x = 6$, the third side $= 24 - 2 \times 6 = 12$

The maximum area has the dimensions 6 m $\times$ 12 m. **[1 mark]**

[6 marks]

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